Colorization Final Exam – Model Comparison Report

1. Overview

In this project, we implemented two distinct colorization approaches on the STL-10 dataset and compared their strengths and weaknesses. **Model 1** is a lightweight four-layer encoder–decoder (LiteColorizer) trained on 128×128 inputs using only an MSE loss. **Model 2** enhances this approach by incorporating semantic features extracted from a frozen, pretrained ResNet-18 encoder, followed by a coarse upsampling stage and a pixelwise LSTM-based refinement.

Both models were trained and evaluated on STL-10, and their performance was measured using MSE, PSNR, and SSIM. Additionally, we visually inspected the colorization results on ten representative test images covering diverse scene types.

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2. Analysis of Model 1

Model 1 achieved rapid convergence and reasonable color assignments but suffered from significant limitations. Its compact architecture allowed training on **5 000 examples** reaching validation losses around **0.0155** (MSE), PSNR of approximately **27 dB**, and SSIM of **0.82**. Despite these respectable numbers, several factors curtailed its ultimate performance:

- **Drastic resolution reduction.** The original design contemplated 512×512 inputs, but practical memory and compute constraints forced us to downsample images to 128×128. Reading 512×512 images from disk was prohibitively slow, and loading such high-resolution data into RAM exceeded available memory. This reduction eliminated many high-frequency details and fine textures essential for photorealistic colorization.

- **Insufficient depth.** With only four convolutional layers in the encoder, the network lacked the capacity to learn complex color distributions, particularly in regions with subtle hue variations or intricate patterns.

- **Mismatch with data resolution.** The STL-10 dataset provides images at 96×96, yet Model 1 expected 128×128. The necessary upscaling and downscaling introduced interpolation artifacts and inconsistent scaling of spatial features.

Nevertheless, Model 1 demonstrated stable training behavior, quickly learned to assign plausible base colors to large regions (e.g., sky, road, foliage), and served as an effective proof of concept.

3. Analysis of Model 2

Model 2 builds upon semantic feature extraction and sequential pixel refinement. By freezing a pretrained ResNet-18 (up to layer 3) as the encoder, we leveraged high-level object and scene understanding without additional training cost. The encoder output was upsampled through four transposed convolution layers to generate a 96×96 coarse color prediction. Finally, a sliding-window LSTM processed local patches (5×5) in a row-major fashion, adding residual corrections to enhance local color coherence.

Training Model 2 for **10 epochs** required approximately **2.5 hours**, yielding a final validation loss of **0.0069**, PSNR of **22.7 dB**, and SSIM of **0.85**. The semantic encoder significantly improved structural consistency and semantic color choices, and the LSTM refinement reduced boundary artifacts compared to Model 1. However, the sequential nature of the LSTM made training and inference extremely slow (about 15 minutes per epoch), and the limited input resolution (96×96) still prevented recovery of fine details.

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4. Conclusion

Model 1 provides a fast, lightweight baseline that learns coarse semantic color mapping with minimal resources. Model 2, by incorporating pretrained semantic features and pixelwise refinement, improves perceptual quality but at the expense of drastically increased computational cost and slower throughput. Future work should focus on integrating multi-scale CNN refinements or generative losses (e.g., perceptual or adversarial) to combine the strengths of both approaches while maintaining practical training times.

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